Miao Li

PhD Student, Georgia Institute of Technology
mli746@gatech.edu | Google Scholar Profile | Personal Website | LinkedIn

Education

Georgia Institute of Technology — Atlanta, GA
Ph.D. in Machine Learning (ISyE) — *Aug. 2023 – May 2027 (expected)*

University of Chicago — Chicago, IL
M.S. in Computational and Applied Mathematics — *2021*

University of California, San Diego — La Jolla, CA
B.S. in Mathematics — *2018*

Research Interests

My research focuses on **efficient LLM inference (particularly for diffusion language models)**, **data-centric post-training**, and **agentic AI systems**. I develop parallel decoding algorithms and end-to-end post-training pipelines that combine self-distillation-based data generation with learner-compatible supervision. I also develop agent harnesses that integrate tool orchestration, structured memory, and verifiable feedback to guide iterative policy discovery and ensure reliable execution.

Publications & Preprints

`* equal contribution`

- **LLMs & Diffusion Language Models** – parallel decoding, post-training, and interpretability

Plan, Verify and Fill: A Structured Parallel Decoding Approach for Diffusion Language Models. **EMNLP** 2026
Miao Li*, Hanyang Jiang*, et al. ★ Spotlight `Non-AR LLM @ COLM 2026`

Match the Canvas, Not Just the Answer: Predictability-Aware SFT for Diffusion Language Models. Submitted to ICLR 2027
Miao Li*, Hanyang Jiang*, et al.

Target-Token Asymmetry Can Masquerade as Depth Specialization. Submitted to ICLR 2027
Danlin Li, **Miao Li**, et al. `+ Accepted at Interp4Discovery @ NeurIPS 2026`

- **Agentic AI** – agent harnesses, RSI, agent benchmarks

OPTAgent: From Standalone Solvers to Deployment-Aware Solver Ecosystems. **MLxOR @ NeurIPS** 2026
Miao Li*, P. Van Hentenryck, et al. `full paper in prep. for ICML 2027`

SkillsBench: Benchmarking How Well Agent Skills Work Across Diverse Tasks. **NeurIPS** 2026
Xiangyi Li, . . . , **Miao Li**, et al. `Datasets & Benchmarks Track`

- **Trustworthy AI & Numerical Optimization** – conformal prediction, optimization surrogates, SQP, federated learning

Constraint-Informed Active Learning for End-to-End ACOPF Optimization Proxies. **EPSR** 2027
Miao Li, M. Klamkin, P. Van Hentenryck, R. Bent, W. Li. `Electric Power Systems Research`

Tightening Optimality Gap with Confidence through Conformal Prediction. **Trustworthy AI @ ICLR** 2026
Miao Li, M. Klamkin, R. Bent, P. Van Hentenryck.

Enabling End-to-End Secure Federated Learning in Biomedical Research on Heterogeneous Computing Environments with APPFLx. **CSBJ** 2025
T.H. Hoang, J. Fuhrman, M. Klarqvist, **Miao Li**, et al. `Comput. Struct. Biotechnol. J., Vol. 28`

A Theoretically Sound Sequential Quadratic Programming Algorithm on Riemannian Manifolds. Preprint
Miao Li, S. Na, M. Kolar. `available upon request`

Works in Progress

OPTAgent-Evolve: Towards Deployment-Aware Optimization Ecosystems. — In preparation for ICML 2027
Miao Li*, Hanyang Jiang*, Changkun Guan*, P. Van Hentenryck, et al.

TextPredOpt: Benchmarking Language Models for Predict-then-Optimize from Text. — In preparation for ICML 2027
Joe Ye, **Miao Li**, P. Van Hentenryck, et al.

EvoDecomp: Agent-Evolved Decomposition for Large-Scale Freight Routing. — In preparation for ICML 2027
Sikai Cheng, **Miao Li**, et al.

Research Experience

Evaluation-Guided Policy Discovery — OPTAgent

- Built an agent harness that evolves solver-orchestration policies through role-based search, policy memory, and verified evaluation feedback.
- Compiled discovered policies into deterministic controllers with enforced compute budgets and no deployment-time LLM calls.
- Increased mean feasible-instance count by **11%** over a direct LLM agent under a shared workload budget.

Efficient Parallel Decoding — Plan-Verify-Fill

- Developed PVF, combining semantic-anchor planning and selective verification to reduce model forward passes by up to **65%** versus confidence-based parallel decoding.

Data-Centric LLM Post-Training — PACS

- Built an end-to-end post-training pipeline spanning solution-hinted **self-distillation**, LLM-based quality filtering, and fine-tuning.
- Extended **distribution-compatible fine-tuning from answers to canvases**, exploiting masked diffusion LLMs' pre-specified, attention-visible canvas—a degree of freedom absent from standard autoregressive teacher forcing—to better match supervision to the model's existing predictive distribution **by adapting prediction conditions**.
- Developed PACS to select training contexts by model predictability—adapting how reference answers are taught while preserving their content.
- Improved benchmark accuracy by up to **12.84** percentage points over vanilla SFT in evaluations spanning LLaDA-8B and Dream-7B.

LLM Interpretability — Target-Token Asymmetry

- Co-authored Target-Token Asymmetry: layer-ablation "depth specialization" can be a target-token artifact (41-condition audit); in GSM8K vs. Wikitext, removing whitespace flips its sign.

Optimality Assessment via Conformal Prediction

- Developed CPUL–OMLT, a lightweight, plug-and-play conformal framework compatible with **any** optimizer providing primal–dual bounds, delivering finite-sample coverage guarantees without modifying existing solvers.
- Achieved up to **3.8× narrower optimality intervals** at 90% target coverage on large-scale economic dispatch, enabling risk-aware stopping decisions and **smarter compute allocation**.

Structure-Aware Active Learning for Optimization Proxies

- Developed constraint-guided active learning that combines learned active-set predictions with validation feedback to target hard, underrepresented solution regimes.
- Reduced **tail prediction errors** on large-scale ACOPF benchmarks (up to 1,354 buses), reaching 90% of converged performance **2–4 rounds earlier** and saving **600–1,200 expensive solver labels**.

Benchmark Contribution

SkillsBench / BenchFlow: GPU Cluster Online Scheduling Task

- Contributed the `gpu-cluster-online-scheduling` task, a verifier-based benchmark module for evaluating agent behavior on online GPU-cluster scheduling under partial information.
- Designed the task around operations-relevant constraints including GPU-type compatibility, capacity limits, CPU/memory feasibility, deadline penalties, fragmentation cost, and active-machine usage.
- In task-level paired evaluation, the curated Skill improved a GPT-5.5-based agent's pass rate from 20% to 40%, highlighting how domain-specific procedural guidance can improve agent performance beyond the base harness.

Work Experience

Argonne National Laboratory Lemont, IL

Research Intern (Mathematics and Computer Science Division) *Oct. 2022 – Aug. 2023*

- Developed `APPFLx`, a secure federated learning framework for biomedical research, enabling privacy-preserving training across heterogeneous computing environments.
- Investigated gradient inversion attacks on biomedical images to strengthen privacy protocols against adversarial reconstruction. *Advisor: Mihai Anitescu*

University of Chicago, Booth School of Business Chicago, IL

Research Assistant *July 2021 – Sept. 2022*

- Conducted theoretical analysis of Sequential Quadratic Programming (SQP) algorithms for optimization on Riemannian manifolds. *Advisor: Mladen Kolar*

Northeast Securities Shanghai, China

Quantitative Research *Nov. 2018 – Aug. 2019*

- Evaluated and backtested multi-factor equity signals for quantitative investment strategies.
- Validated factor stability, portfolio construction assumptions, and risk-adjusted performance across historical market regimes.

Teaching Experience

- **Teaching Assistant**, UC San Diego (2016 – 2018): Game Theory, Calculus I/II, Intro to Analysis, Nonlinear Dynamics.

Honors & Awards

- **George Nemhauser Fellowship** (Georgia Tech)